

1. Rollout

π_t



Sample transitions

Rollout + GAE

$\hat{A}_r, \hat{A}_{c_k}$

2. Flow inputs



Clustered states
Transition flow

$\hat{F}, A, C_k$

3. Constraints

Soft budgets

$$\langle C_k, F \rangle \leq b_k$$



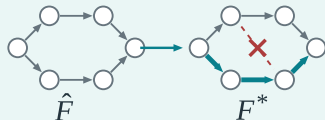
$$F_H = 0$$

Hard exclusions

4. SafeOT projection

$$\min_{F \geq 0} -\langle A, F \rangle + \epsilon \text{KL}(F \| \hat{F})$$

s.t. flow balance + constraints



Target flow is not executed

advantages

$\bar{\lambda}_k^*$

clip + EMA

λ_k^*

budget duals

5. Priced update

Observed episode costs

$$\beta_k \leftarrow [\beta_k + \eta(\bar{J}_k - B_k)]_+ \rightarrow \oplus$$

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

$$\tilde{A} = \frac{\hat{A}_r - \sum_k \lambda_k \hat{A}_{c_k}}{1 + \sum_k \lambda_k}$$

PPO/TRPO $\rightarrow \pi_{t+1}$

next rollout